

CS 105: Department Introductory Course on Discrete Structures

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Lecture 09 – Basic structures: Relations

Summary and moving on...

Week 01, 02 and 03: Proofs and Reasoning

- ▶ Propositions, Predicates, Quantifiers
- ▶ Theorems and Types of Proofs
- ▶ Induction and variants

Week 04 and 05: Basic Mathematical Structures

- ▶ Finite and infinite sets.
- ▶ Using functions to compare sets: focus on bijections.
- ▶ Countable, countably infinite and uncountable sets.
- ▶ Cantor's diagonalization argument (A new powerful proof technique!).

Next: Basic Mathematical Structures – Relations

Relations

Definition: Function

Let A, B be two sets. A **function** f from A to B is a subset R of $A \times B$ such that

- (i) $\forall a \in A, \exists b \in B$ such that $(a, b) \in R$, and
 - (ii) if $(a, b) \in R$ and $(a, c) \in R$, then $b = c$.
- Now, suppose A is the set of all Btech students and B is the set of all courses. Clearly, we can assign to each student the courses he/she is taking. Is this a function?
 - By removing the two extra assumptions in the defn, we get:

Definition: Relation

- A **relation** R from A to B is a subset of $A \times B$. If $(a, b) \in R$, we also write this as aRb .
- Thus, a relation is a way to relate the elements of two (not necessarily different) sets.

Examples and representations of relations

We write $R(A, B)$ for a relation from A to B and just $R(A)$ if $A = B$. Also if A is clear from context, we just write R .

Examples of relations

- ▶ All functions are relations.
- ▶ $R_1(\mathbb{Z}) = \{(a, b) \mid a, b \in \mathbb{Z}, a - b \text{ is even} \}$.
- ▶ $R_2(\mathbb{Z}) = \{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$.
- ▶ Let S be a set, $R_3(\mathcal{P}(S)) = \{(A, B) \mid A, B \subseteq S, A \subseteq B\}$.
- ▶ Relational databases are practical examples.

Representations of a relation from A to B .

- ▶ As a set of **ordered pairs of elements**, i.e., subset of $A \times B$.
- ▶ As a **directed graph**.
- ▶ As a **(database) table**.

Use of relations

Practical application in relational databases: IMDB, university records, etc.

But, why study relations in this course?

- ▶ Functions were special kinds of relations that were useful to compare sets.
- ▶ Are there other **special** relations? What are they useful for?
 - ▶ Equivalence relations
 - ▶ Partial orders

Partitions of a set – grouping “like” elements

Examples

- ▶ Natural numbers are partitioned into even and odd.
- ▶ This class is partitioned into sets of students from **same** hostel.

How do you define a partition?

Definition

A partition of a set S is a **set P of its subsets** such that

- ▶ if $S' \in P$, then $S' \neq \emptyset$.
- ▶ $\bigcup_{S' \in P} S' = S$: its union covers entire set S .
- ▶ If $S_1, S_2 \in P$, then $S_1 \cap S_2 = \emptyset$: sets are disjoint.

Can you think of two trivial partitions that any (non-empty) set must have? (They may coincide for singleton sets)

Interpreting partitions as relations

Thus, a partition divides S into subsets that each contain (only) elements that share some common property. e.g.,

- ▶ Evenness or oddness (formally, remainder modulo 2!).
- ▶ Same hostel...
- ▶ But, this sounds like relation, right? Which one?

Relation generated by a partition

- ▶ Clearly all elements in a set of the partition are related by the “sameness” or “likeness” property.
- ▶ So can we define this as a relation? aRb if a is “like” b .
- ▶ Formally, we define $R(S)$ by aRb if a and b belong to the same set in the partition of S .

What properties does this relation have?

Properties of a relation generated by a partition

1. All elements must be related to themselves
 - ▶ A relation $R(S)$ is called **reflexive** if for all $a \in S$, aRa .
2. If a is “like” b , then b must be “like” a .
 - ▶ A relation $R(S)$ is called **symmetric** if for all $a, b \in S$, we have aRb implies bRa .
3. If a is “like” b and b is “like” c , then a must be “like” c .
 - ▶ A relation $R(S)$ called **transitive** if for all $a, b, c \in S$, we have aRb and bRc implies aRc .

Any other defining properties?

Definition

A relation which satisfies all these three properties is called an **equivalence relation**.

Thus, from any **partition**, we get an **equivalence relation**.

Examples

- **Reflexive:** $\forall a \in S, aRa$.
- **Symmetric:** $\forall a, b \in S, aRb$ implies bRa .
- **Transitive:** $\forall a, b, c \in S, aRb, bRc$ implies aRc .
- **Equivalence:** Reflexive, Symmetric and Transitive.

Relation	Refl.	Sym.	Trans.	Equiv.
aR_4b if students a and b take same set of courses	✓	✓	✓	✓
aR_5b if student a takes course b				
$\{(a, b) \mid a, b \in \mathbb{Z}, (a - b) \bmod 2 = 0\}$				
$\{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$				
$\{(a, b) \mid a, b \in \mathbb{Z}, a < b\}$				
$\{(a, b) \mid a, b \in \mathbb{Z} \setminus \{0\}, a \mid b\}$				
$\{(a, b) \mid a, b \in \mathbb{R}, a - b < 1\}$				
$\{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}), (ad = bc)\}$				

Partitions of a set – grouping “like” elements

Definition

A partition of a set S is a set P of its subsets such that

- ▶ if $S' \in P$, then $S' \neq \emptyset$.
- ▶ $\bigcup_{S' \in P} S' = S$: its union covers entire set S .
- ▶ If $S_1, S_2 \in P$, then $S_1 \cap S_2 = \emptyset$: sets are disjoint.

Example: natural numbers partitioned into even and odd...

Theorem

Every partition of set S gives rise to a **canonical** equivalence relation R on S , namely,

- ▶ aRb if a and b belong to the same set in the partition of S .

Is the converse true? Can we generate a partition from every equivalence relation?

Equivalence classes

Definition

- ▶ Let R be an equivalence relation on set S , and let $a \in S$.
- ▶ Then the **equivalence class** of a , denoted $[a]$, is the set of all elements related to it, i.e., $[a] = \{b \in S \mid (a, b) \in R\} = \{b \in S \mid aRb\}$.

In $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid (a - b) \bmod 5 = 0\}$, what are $[0]$, $[1]$?

Lemma

Let R be an equivalence relation on S . Let $a, b \in S$. Then, the following statements are equivalent:

1. aRb
2. $[a] = [b]$
3. $[a] \cap [b] \neq \emptyset$.

Proof Sketch: (1) to (2) symm and trans, (2) to (3) refl, (3) to (1) symm and trans. (H.W.: Redo the proof formally.)

From equivalence relations to partitions

Theorem

1. Let R be an equivalence relation on S . Then, the equivalence classes of R form a partition of S .
2. Conversely, given a partition P of S , there is an equivalence relation R whose equivalence classes are exactly the sets of P .

Proof sketch of (1): Union, non-emptiness follows from reflexivity. The rest (pairwise disjointness) follows from the previous lemma.

(H.W.): Write the formal proofs of (1) and (2).

More “applications” of equivalence relations

Defining new objects using equivalence relations

Consider $R = \{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}), (ad = bc)\}$.

- ▶ Then the equivalence classes of R define the rational numbers.
- ▶ e.g., $[\frac{1}{2}] = [\frac{2}{4}]$ are two names for the same rational number.
- ▶ Indeed, when we write $\frac{p}{q}$ we implicitly mean $[\frac{p}{q}]$.
- ▶ With this definition, why are addition and multiplication “well-defined”?

Can we define **integers** and **real numbers** starting from naturals by using equivalence classes?

Geometrical objects using equivalence relations

Cut-and-paste

Consider the relation $R([0, 1]) = \{aRb \mid a, b \in [0, 1], \text{ either } a = b \text{ or } a = 1, b = 0, \text{ or } a = 0, b = 1\}$.

- ▶ Is R an equivalence relation? What does it define?
- ▶ This is $[0, 1]$ in which the end-points have been related to each other.
- ▶ So the equivalence classes form a “loop”, since end-points are joined. If we imagine $[0, 1]$ as a 1-length string, we have glued its ends!

Geometrical objects using equivalence relations

Forming 2D objects

Consider a rectangular piece of the real plane, $[0, 1] \times [0, 1]$.

- ▶ Define $R_1([0, 1] \times [0, 1])$ by $(a, b)R_1(c, d)$ if
 - ▶ $(a, b) = (c, d)$ or
 - ▶ $b = d, a = 0, c = 1$ or
 - ▶ $b = d, c = 0, a = 1$.

Is R_1 an equivalence relation? What do its equivalence classes define?

- ▶ Define $R_2([0, 1] \times [0, 1])$ by $(a, b)R_2(c, d)$ if
 - ▶ $(a, b) = (c, d)$ or
 - ▶ $a, b, c, d \in \{0, 1\}$.

Is R_2 an equivalence relation? What does it define?

Can you build even more interesting “shapes”? Torus? Mobius strip?!